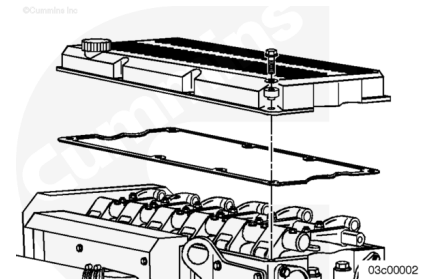


Trainings ▾

Preparatory Steps

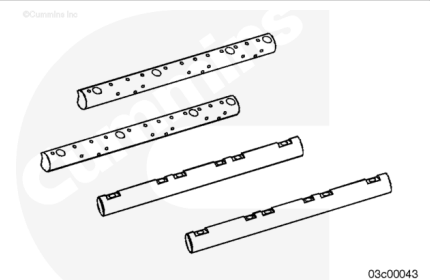
- Remove the engine brake solenoid valve wiring harness.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3
(/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html)
- Remove the engine brake wiring harness. Refer to Procedure 020-015 in Section 20
(/qs3/pubsys2/xml/en/procedures/10/10-020-015.html)



Remove

There are four rocker lever shafts:

- Front - valve and brake rocker lever shaft
- Rear - valve and brake rocker lever shaft
- Front - injector rocker lever shaft
- Rear - injector rocker lever shaft.

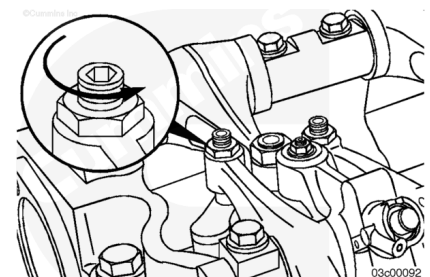


⚠ CAUTION ⚠

Some valves will not be loose. Do not tighten or bottom adjusting screw against lever. Engine damage can occur.

Loosen the valve and injector adjusting screw locknuts on each rocker lever.

Turn the adjusting screws **counterclockwise** until the rocker levers are loose.



⚠ CAUTION ⚠

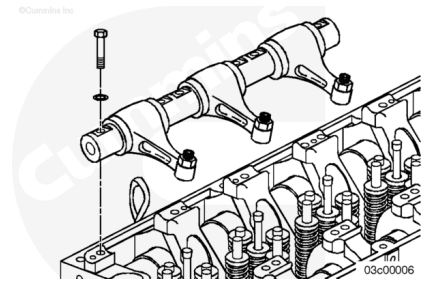
Use care when handling shafts. Rocker levers can fall off the shaft during removal and damage the levers.

Remove the valve and brake rocker lever shaft capscrews, and the rocker lever shaft assemblies.

Remove the injector, rocker lever shaft capscrews, and the rocker lever shaft assemblies.

Remove the valve crossheads.

To prevent increased wear, mark each crosshead and rocker lever shaft assembly as it is removed so it can be reinstalled back in its original location.

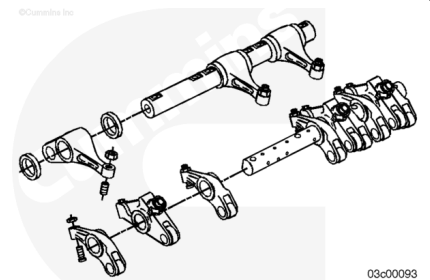


Remove the rocker levers and brake levers from the shafts.

Remove the rocker lever spacers from the shaft.

Remove the locknuts and the adjusting screws from the rocker levers.

To prevent increased wear, mark each adjusting screw as it is removed so it can be reinstalled back in its original rocker lever.



Clean and Inspect for Reuse

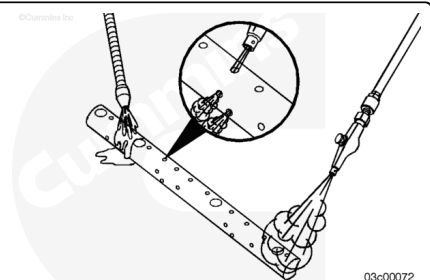
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠



Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

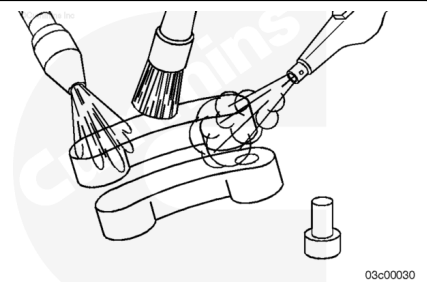
Use steam or solvent to clean the rocker lever shaft.

Dry with compressed air.

Be sure to blow out the oil passages.

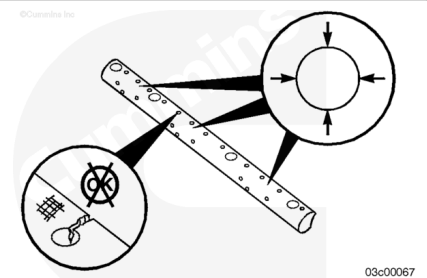
Use steam or solvent to clean the valve crossheads, rocker lever adjusting screw, and nuts.

Dry with compressed air.



Inspect the valve and brake rocker lever shafts for cracks or other damage.

Measure the valve and brake rocker lever shaft outside diameters in the bushing wear area.



Valve and Brake Rocker Lever Shaft Outside Diameter

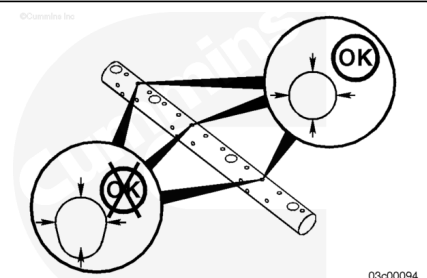
mm		in
34.950	MIN	1.3760
35.000	MAX	1.3780

If the valve or brake rocker lever shaft is **not** within specifications, it **must** be replaced.

Inspect the valve and brake rocker lever shafts for wear in the brake detent pockets.

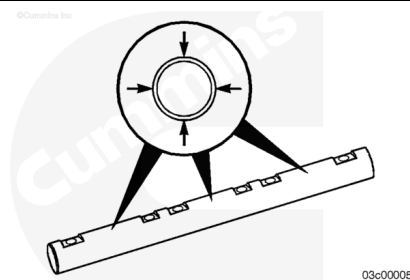
A raised edge on one side of the detent pocket is normal. The shaft does **not** need to be replaced.

An elongated detent pocket, typically toward the camshaft follower end of the brake rocker lever, indicates excessive wear and the shaft **must** be replaced.



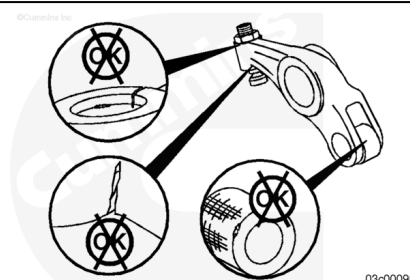
Measure the injector rocker lever shaft outside diameter in the bushing wear area.

Injector Rocker Lever Shaft Outside Diameter		
mm		in
44.950	MIN	1.7697
45.000	MAX	1.7717



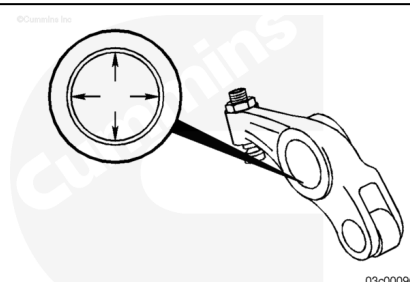
If the rocker lever shaft is **not** within specifications, it **must** be replaced.

Inspect the rocker levers for cracks or unusual wear and thread condition.



Measure the valve and brake rocker lever bushing bore inside diameter.

Valve and Brake Rocker Lever Bushing Bore Inside Diameter		
mm		in
35.037	MIN	1.3794
35.138	MAX	1.3834



If the valve or brake rocker lever bushing is **not** within specifications, the rocker lever **must** be replaced.

If the valve camshaft and/or the rocker levers need to be replaced, use the Camshaft Compatibility Matrix to determine whether the camshaft and rocker lever roller combination being installed is compatible.

If the matrix states that the combination of the camshaft and rocker lever roller is "No", the correct combination **must** be installed using the Valve Camshaft Compatibility Matrix.

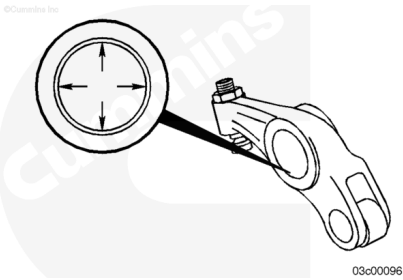
The wide rocker lever roller is 17 mm [0.67 in] and the narrow rocker lever roller is 15 mm [0.59 in].

The wide camshaft lobe is 20 mm [0.79 in] and the narrow camshaft lobe is 18.5 mm [0.73 in].

Valve Camshaft Compatibility Matrix	Wide Camshaft Lobe	Narrow Camshaft Lobe
Wide Rocker Lever Roller	Yes	No
Narrow Rocker Lever Roller	Yes	Yes

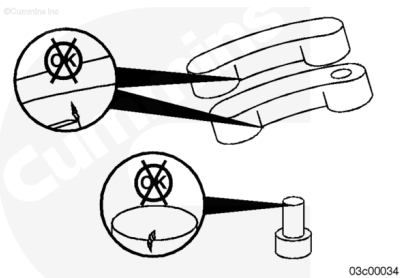
Measure the injector rocker lever bushing bore inside diameter.

Injector Rocker Lever Bushing Bore Inside Diameter		
mm		in
45.037	MIN	1.7731
45.138	MAX	1.7771



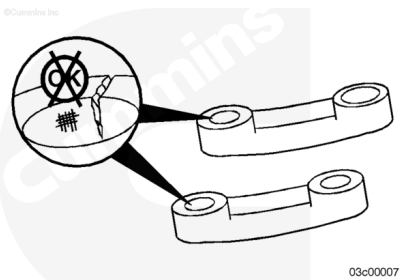
If the injector rocker lever bushing is **not** within specifications, the rocker lever **must** be replaced.

Inspect the valve crossheads and crosshead guide pins for damage.



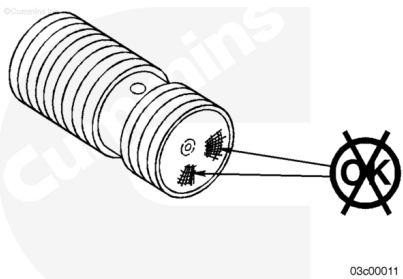
The contact area on the crossheads **must** be smooth with an even seating pattern.

If any distortions or cracks are found on the valve crossheads or adjusting screws, they **must** be replaced.



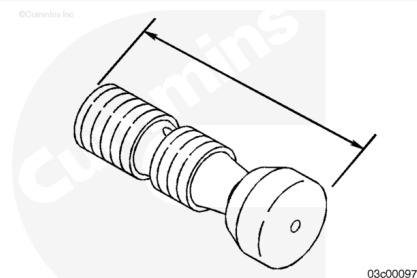
Inspect the injector rocker lever adjusting screw socket for excessive wear.

If excessive wear is found, the injector rocker lever adjusting screw **must** be replaced.



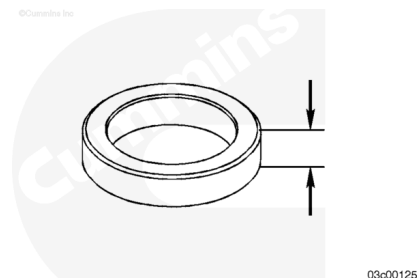
Measure the intake and exhaust valve adjusting screws for wear. Make sure the adjusting screw is perpendicular for measurement.

Intake and Exhaust Valve Adjusting Screw Length		
mm		in
46	MIN	1.81



Measure the rocker lever spacers for wear.

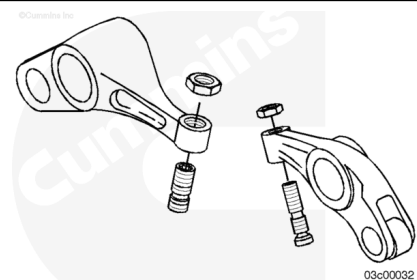
Rocker Lever Spacer Width		
mm		in
11.5	MIN	0.453



If the rocker lever spacer is **not** within specification, it **must** be replaced.

Install

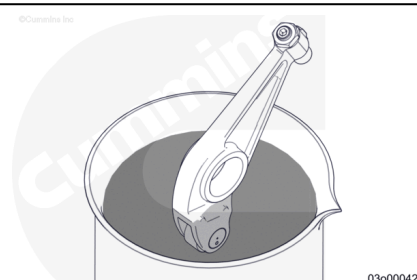
Install the adjusting screw and locknut into each rocker lever. Do **not** tighten the locknuts.



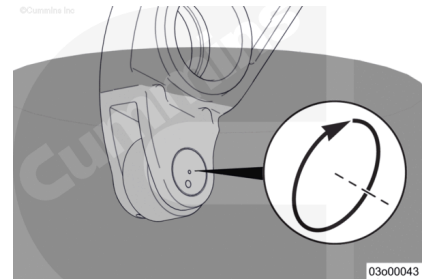
⚠ CAUTION ⚠

Both new and used rocker levers must be lubricated as directed. Failure to lubricate the rocker lever as directed can cause extensive engine damage.

Immerse the roller end of each rocker lever in a container of clean engine lubricating oil.

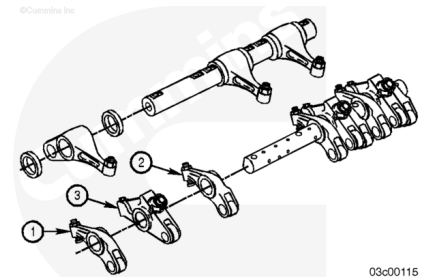


While still immersed, rotate the roller around the roller pin at least once to make sure the joint is adequately lubricated.



Install the injector rocker levers on the shaft.

Install the exhaust (1), intake (2), and engine brake (3) rocker levers onto the shaft.

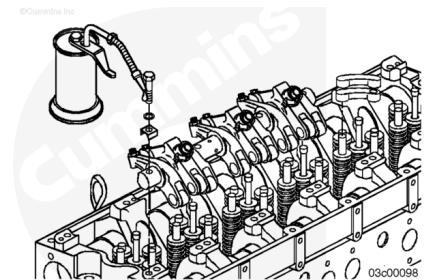


⚠ WARNING ⚠

Clear the rocker lever assembly mounting capscrew holes of oil and debris. Extensive engine damage can result if this warning instruction is not followed.

Install the valve crossheads.

Position the valve/brake rocker lever assemblies onto the cylinder head.

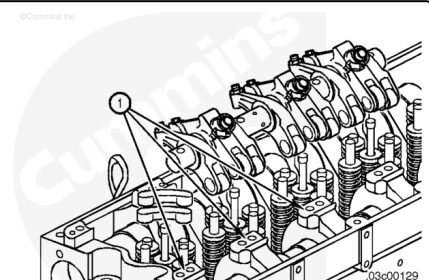


⚠ CAUTION ⚠

To reduce the possibility of valve rocker shaft damage due to inadequate brake rocker lever running clearance, follow the specified procedure to seat the valve rocker shafts on the cylinder head pedestals (1).

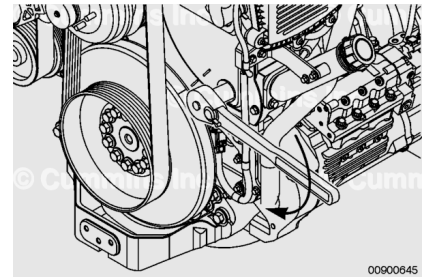
Each valve rocker shaft **must** be seated in the cylinder head pedestals prior to tightening the rocker shaft capscrews.

The front and rear rocker lever shaft assemblies will seat separately as the engine is rotated in the **clockwise** direction.



Use the compressor drive or barring device to bar the engine over in the direction of rotation, **clockwise** as viewed from the front of the engine.

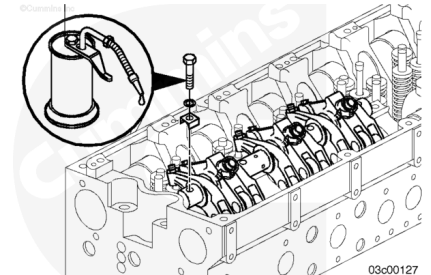
Align the A mark on the vibration damper with the timing mark on the gear cover.



Either the front or the rear rocker shaft assembly will be fully seated on the cylinder head pedestals at this point.

Check the position of the front and rear valve rocker shaft assemblies to determine which one is fully seated.

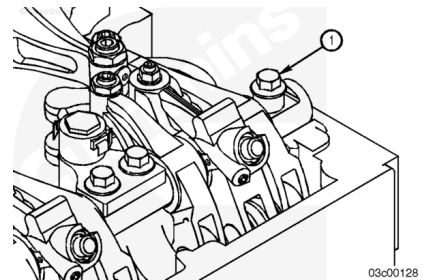
Lubricate the capscrews of the fully seated rocker shaft with clean engine oil.



⚠ CAUTION ⚠

To reduce the possibility of valve rocker shaft damage because of inadequate brake rocker lever running clearance, follow the specified capscrew torque procedure. The valve rocker lever shaft has a machined flat on the front of the shaft to provide proper brake rocker lever position.

Therefore, the front rocker shaft capscrew (1) must be tightened first to control brake rocker lever position. Brake lever running clearance must always be checked after valve rocker lever shaft installation.



Check that the rocker lever locking nuts are loose and adjusting screws are positioned for maximum clearance before proceeding.

Tighten the front mounting capscrew for the fully seated valve rocker lever shaft to the initial torque value to locate the shaft.

Torque Value: 30 N·m [22 ft-lb]

Tighten all the capscrews to the final torque value using the torque plus angle method. Tighten the capscrews from the center out.

Torque Value:

1. 30 N·m [22 ft-lb]
2. Rotate 60 degrees.

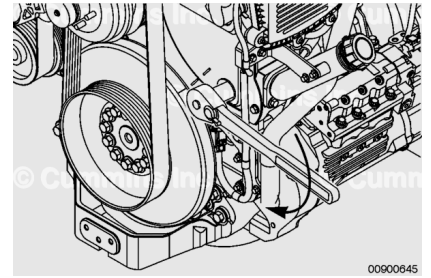
Check the brake running clearance. Refer to Procedure 003-004 in Section 3 (/qs3/pubsys2/xml/en/procedures/10/10-003-004-tr.html).

Use the compressor drive or barring device to bar the engine over in the direction of rotation, **clockwise** as viewed from the front of the engine.

Align the B mark on the vibration damper with the timing mark on the gear cover.

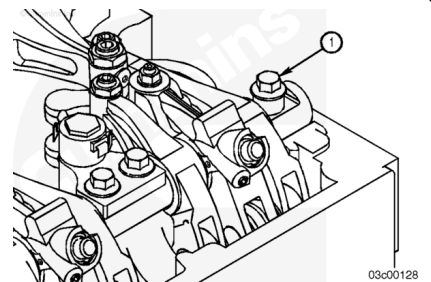
The remaining rocker shaft will be fully seated at this point.

Lubricate the capscrews of the fully seated rocker shaft with clean engine oil.



⚠ CAUTION ⚠

To reduce the possibility of valve rocker shaft damage because of inadequate brake rocker lever running clearance, follow the specified capscrew torque procedure. The valve rocker lever shaft has a machined flat on the front of the shaft to provide proper brake rocker lever position. Therefore, the front rocker shaft capscrew must be tightened first to control brake rocker lever position. Brake lever running clearance must always be checked after valve rocker lever shaft installation.



Check that the rocker lever locking nuts are loose and adjusting screws are positioned for maximum clearance before proceeding.

Tighten the front mounting capscrew for the remaining valve rocker lever shaft to the initial torque value to locate the shaft.

Torque Value: 30 n•m [22 ft-lb]

Tighten all the capscrews to the final torque value using the torque plus angle method. Tighten the capscrews from the center out.

Torque Value:

1. 30 n•m [22 ft-lb]
2. Rotate 60 degrees.

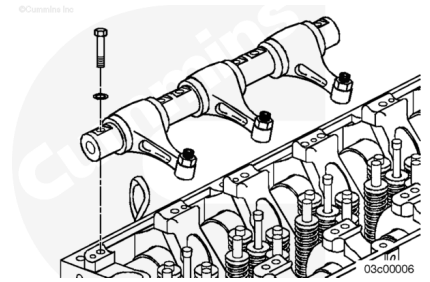
Check the brake running clearance. Refer to Procedure 003-004 in Section 3. (/qs3/pubsys2/xml/en/procedures/10/10-003-004-tr.html)

Position the injector rocker lever assemblies onto the cylinder head.

Lubricate the injector rocker lever shaft capscrews with clean engine oil. Install the capscrews and washers.

Tighten all the mounting capscrews.

Torque Value: 68 n•m [50 ft-lb]



Loosen the capscrews one at a time 360° and tighten again, using the torque-plus-angle method. Tighten the capscrews from the center out.

Torque Value:

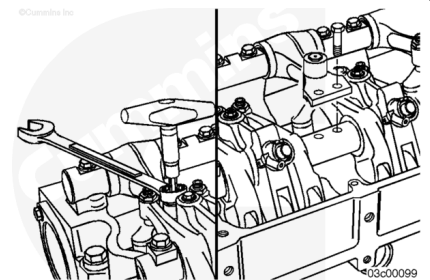
1. 30 n•m [22 ft-lb]

2. Rotate 60 degrees

Adjust the valves, injectors, and brakes.

Refer to Procedure 003-004

(</qs3/pubsys2/xml/en/procedures/10/10-003-004-tr.html>) in Section 3.



Finishing Steps

- Install the engine brake wiring harness. Refer to Procedure 020-015 in Section 20 (</qs3/pubsys2/xml/en/procedures/10/10-020-015.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>)
- Install the engine brake solenoid valve wiring harness.

